Unit 1

Module 1 :

Matrices

1

Matrices

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Syllabus

Linear Independence; Row Echelon form (REF) and Reduced Row Echelon form (RREF) of a Matrix, Rank of a Matrix using REF/RREF; Inverse of a matrix using Gauss-Jordan method; Solution of System of linear equations by elementary row operations; Symmetric, skew-symmetric and orthogonal matrices; Eigen values and eigenvectors; Diagonalization of matrices; Inverse of a Matrix by Cayley-Hamilton Theorem.

1.0 Definition of Matrix

- **Definition of matrix :** If m and n are positive integers, then an $m \times n$ **matrix** is a rectangular array

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \text{m rows}$$

n columns

- In which each **entry**, a_{ij}, of the matrix is a number. An $m \times n$ matrix (read "m by n") has m **rows** (horizontal lines) and n **columns** (vertical lines).
- The entry a_{ij} is located in the i^{th} row and the j^{th} column. The index i is called the **row subscript** because it identifies the row in which the entry lies and the index j is called the **column subscript** because it identified the column in which the entry lies.
- A matrix with m rows and n columns (an $m \times n$ matrix) is said to be of **size** $m \times n$. If $m \times n$, the matrix is called **square** of **order** n. For a square matrix, the entries $a_{11}, a_{22}, a_{33}, \ldots$ are called the **main diagonal** entries.

Examples of matrices :

- Each matrix has the indicated size.

a) Size : 1×1 $\quad [2]$

b) Size : 2×2 $\quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

c) Size : 1×4 $\quad \begin{bmatrix} 1 & -3 & 0 & \frac{1}{2} \end{bmatrix}$

d) Size : 3×2 $\quad \begin{bmatrix} e & \pi \\ 2 & \sqrt{2} \\ -7 & 4 \end{bmatrix}$

1.1 Elementary Row Operations in Matrix

- Row reduction is a process by which a matrix is simplified into an "equivalent" matrix which is easier to use overall. In order to make this procedure more canonical, we'll perform our reduction using a very precise collection of operations known as elementary row operations.

Throughout, we'll refer to the matrix

$$M = \begin{bmatrix} 4 & 1 & -2 & 0 \\ 1 & 0 & 0 & -5 \\ 0 & 1 & 3 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix}$$

for all of our examples.

- There are three elementary row operations that we can perform on a matrix to get a new matrix which is considered "row equivalent" to it : 1.
- Interchange two rows.

For example

1. Interchanging rows 2 and 4 (shorthand : $R_2 \leftrightarrow R_4$) in M yields the following :

$$\begin{bmatrix} 4 & 1 & -2 & 0 \\ 1 & 0 & 0 & -5 \\ 0 & 1 & 3 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{bmatrix} 4 & 1 & -2 & 0 \\ 1 & 1 & -1 & -1 \\ 0 & 1 & 3 & -1 \\ 1 & 0 & 0 & -5 \end{bmatrix}$$

Multiply all entries in a row by a nonzero constant.

2. We can multiply row 1 of M by 3 (shorthand : $R_1 \leftrightarrow 3R_1$) to get the following :

$$\begin{bmatrix} 4 & 1 & -2 & 0 \\ 1 & 0 & 0 & -5 \\ 0 & 1 & 3 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow 3R_1} \begin{bmatrix} 12 & 3 & -6 & 0 \\ 1 & 0 & 0 & -5 \\ 0 & 1 & 3 & -1 \\ 1 & 1 & 0 & -1 \end{bmatrix}$$

Add (or subtract) a nonzero multiple of one row to another row.

3. Let's say we wanted to add 4 times row 2 to row 3, i.e. we leave every row the same except row 3 and we change row 3 by adding to it $4R_2$ (shorthand : $R_3 \leftrightarrow R_3 + 4R_2$).

We could do this all at once, but to split it into steps, we could :

a) Compute $4R_2$, i.e. $4.\langle 1, 0, 0, -5 \rangle = \langle 4, 0, 0, -20 \rangle$;

b) Compute R_3 plus $4R_2$,

i.e. $\langle 0, 1, 3, -1 \rangle + \langle 4, 0, 0, -20 \rangle = \langle 4, 1, 3, -21 \rangle$ and

c) Form the new matrix having the same entries as M in rows 1, 2 and 4 and having $R_3 + 4R_2 = \langle 4, 1, 3, -21 \rangle$ as its third row.

Hence, the result is :

$$\begin{bmatrix} 4 & 1 & -2 & 0 \\ 1 & 0 & 0 & -5 \\ 0 & 1 & 3 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_3 + 4R_2} \begin{bmatrix} 4 & 1 & -2 & 0 \\ 1 & 0 & 0 & -5 \\ 4 & 1 & 3 & -21 \\ 1 & 1 & -1 & -1 \end{bmatrix}$$

1.2 Row Echelon and Reduced Row Echelon Forms

Row Echelon form

- **Definition** : A matrix is in Row Echelon Form (REF) if it satisfies the following three properties :
- A matrix in **row-echelon form** has the following properties.

1. All rows consisting entirely of zeros occur at the bottom of the matrix.
2. For each row that does not consist entirely of zeros, the first nonzero entry is 1 (called a **leading 1**).
3. For two successive (nonzero) rows, the leading 1 in the higher row is farther to the left than the leading 1 in the lower row.

- As a remark, note that the entries above the leading 1 of such a matrix may or may not equal 0. For instance, both of the following matrices are in REF :

The matrices below are in row-echelon form.

(a) $\begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & -5 & 2 & -1 & 3 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Reduced row Echelon form

- **Definition** : A matrix is in Reduced Row Echelon Form (RREF) if it satisfies the following three properties :

1. It is in REF;
2. Each leading 1 is the only nonzero entry in its column.

- The matrices shown in parts (b) and (d) are in *reduced* row-echelon form. The matrices listed below are not in row-echelon form.

(e) $\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & -3 \end{bmatrix}$ (f) $\begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & -4 \end{bmatrix}$

1.3 Solution of System of Linear Equations by Gauss Elimination and Gauss Jordan Methods

GTU : Summer-19, 21, Winter-19, 20, 21, 22

- A *linear equation* can be written as $a_1x_1 + a_2x_2 + ... + a_nx_n = b$, where $a_1, ..., a_n$ and b are real numbers known in advance.
- For example, $4x_1 - 5x_2 + 2 = x_1$ is a linear equation, $\sqrt{2x_1} + \pi x_2 = -1$ is also a linear equation, but $x_1 = 2x_2x_4$ and $x_2 = 2\sqrt{x_1} - 6$ are not linear equations.
- A system of *linear equations* (or a *linear system*) is a collection of one or more linear equation involving the same variables. For example

$$\begin{cases} 2x_1 - x_2 + 1.5x_3 = 8 \\ x_1 - 4x_3 = -7 \end{cases}$$

- A solution to a linear system is a list of numbers $(s_1, ..., s_n)$ that make each equation a true statement. The set of all possible solutions is called a *solution* set.
- **Geometric interpretation** : As we have learned in analytic geometry, the solution of a linear system in two variables can have

1) Exactly one solution (when two lines intersect at a unique point)

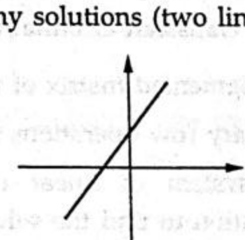

Fig. 1.3.1

2) Infinitely many solutions (two lines coincide)

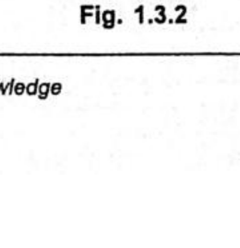

Fig. 1.3.2

3) No solution (two lines are parallel but not the same)

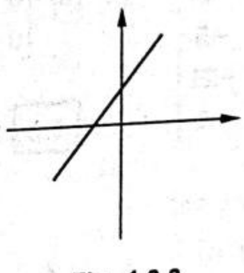

Fig. 1.3.3

Existence and uniqueness questions

1) Does the solution exist ? (Existence question)
2) If a solution exists, is it the only one solution ? that is, is the solution unique or infinite ? (Uniqueness question)

- If a linear system doesn't have a solution, we call the system *inconsistent,* otherwise, the system is called *consistent.*
- One very common use of matrices is to represent systems of linear equations. The matrix derived from the coefficients and constant terms of a system of linear equations is called **the augmented matrix** of the system. The matrix containing only the coefficients of the system is called **the coefficient matrix** of the system. Here is an example.

System	Augmented matrix	Coefficient matrix
$\begin{aligned} x-4y+3z&=5 \\ -x+3y-z&=-3 \\ 2x-4z&=6 \end{aligned}$	$\begin{bmatrix} 1 & -4 & 3 & 5 \\ -1 & 3 & -1 & -3 \\ 2 & 0 & -4 & 6 \end{bmatrix}$	$\begin{bmatrix} 1 & -4 & 3 \\ -1 & 3 & -1 \\ 2 & 0 & -4 \end{bmatrix}$

- When forming either the coefficient matrix or the augmented matrix of a system, you should begin by aligning the variables in the equations vertically.

Givensystem	Align variables	Augmented matrix
$\begin{aligned} x_1+3x_2&=9 \\ -x_2+4x_3&=-2 \\ x_1-5x_3&=0 \end{aligned}$	$\begin{aligned} x_1+3x_2&=9 \\ -x_2+4x_3&=-2 \\ x_1-5x_3&=0 \end{aligned}$	$\begin{bmatrix} 1 & 3 & 0 & 9 \\ 0 & -1 & 4 & -2 \\ 1 & 0 & -5 & 0 \end{bmatrix}$

Solution of system of linear equations by Gauss Elimination method

- Gaussian elimination with back-substitution works well as an algorithmic method for solving systems of linear equations. For this algorithm, the order in which the elementary row operations are performed is important. Move from *left to right by columns,* changing all entries directly below the leading 1's to zeros.

An algorithm for Gaussian elimination

1. Write the augmented matrix of the system of linear equations.
2. Use elementary row operations to rewrite the augmented matrix in row-echelon form.
3. Write the system of linear equations corresponding to the matrix in row-echelon form and use back-substitution to find the solution.

Gaussian elimination with back-substitution

Ex. 1.3.1 : *Solve the system.*

$x_2+x_3-2x_4=-3$

$x_1+2x_2-x_3=2$

$2x_1+4x_2+x_3-3x_4=-2$

$x_1-4x_2-7x_3-x_4=-19$

Sol. : The augmented matrix for this system is

$$\begin{bmatrix} 0 & 1 & 1 & -2 & -3 \\ 1 & 2 & -1 & 0 & 2 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -4 & -1 & -19 \end{bmatrix}$$

Obtain a leading 1 in the upper left corner and zeros elsewhere in the first column.

$$\sim \begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & -1 & -19 \end{bmatrix}$$
$\Leftarrow$ The first two rows $R_1 \leftrightarrow R_2$
$\Leftarrow$ are interchanged.

$$\sim \begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 1 & -4 & -7 & -1 & -19 \end{bmatrix}$$
Adding -2 times the first row to the third row
$\Leftarrow$ produces a new third row. $R_3+(-2)R_1 \rightarrow R_3$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 0 & -6 & -6 & -1 & -21 \end{bmatrix}$$
Adding -1 times the first row to the fourth row $R_4+(-1)R_1 \rightarrow R_4$
$\Leftarrow$ produces a new fourth row.

Now that the first column is in the desired form, you should change the second column as shown below.

$$\sim \begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 0 & 0 & 0 & -13 & -39 \end{bmatrix}$$
Adding 6 times the second row to the fourth row
$\Leftarrow$ produces a new fourth row.
$R_4+(6)R_2 \rightarrow R_4$

To write the third column in proper form, multiply the third row by $\frac{1}{3}$.

$$\sim \begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & -13 & -39 \end{bmatrix}$$
Multiplying the third row by $\frac{1}{3}$
$\Leftarrow$ produces a new third row. $\left(\frac{1}{3}\right)R_3 \rightarrow R_3$

Similarly, to write the fourth column in proper form, you should multiply the fourth row by $-\frac{1}{13}$.

$$\sim \begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$
Multiplying the fourth row by $-\frac{1}{13}$
$\Leftarrow$ produces a new fourth row $\left(-\frac{1}{13}\right)R_4 \rightarrow R_4$

The matrix is now in row-echelon form and the corresponding system of linear equations is as shown below.

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 2 \\ x_2 + x_3 - 2x_4 &= -3 \\ x_3 - x_4 &= -2 \\ x_4 &= 3 \end{aligned}$$

Using back-substitution, you can determine that the solution is $x_1 = -1$, $x_2 = 2$, $x_3 = 1$, $x_4 = 3$.

- **When solving a system of linear equations, remember that it is possible for the system to have no solution. If during the elimination process you obtain a row with all zeros except for the last entry, it is unnecessary to continue the elimination process. You can simply conclude that the system is inconsistent and has no solution.**

A system with no solution

Ex. 1.3.2 : *Solve the system.*

$x_1 - x_2 + 2x_3 = 4$
$x_1 + x_3 = 6$
$2x_1 - 3x_2 + 5x_3 = 4$
$3x_1 + 2x_2 - x_3 = 1.$

Sol. : The augmented matrix for this system is,

$$\begin{bmatrix} 1 & -1 & 2 & 4 \\ 1 & 0 & 1 & 6 \\ 2 & -3 & 5 & 4 \\ 3 & 2 & -1 & 1 \end{bmatrix}$$

Apply Gaussian elimination to the augmented matrix.

$R_2 + (-1)R_1 \to R_2$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 2 & -3 & 5 & 4 \\ 3 & 2 & -1 & 1 \end{bmatrix}$$

$R_3 + (-2)R_1 \to R_3$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & -1 & 1 & -4 \\ 3 & 2 & -1 & 1 \end{bmatrix}$$

$R_4 + (-3)R_1 \to R_4$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & -1 & 1 & -4 \\ 0 & 5 & -7 & -11 \end{bmatrix}$$

$R_3 + R_2 \rightarrow R_3$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & -2 \\ 0 & 5 & -7 & -11 \end{bmatrix}$$

Note that the third row of this matrix consists of all zeros except for the last entry. This means that the original system of linear equations is inconsistent.

A system with infinite solutions

Ex. 1.3.3 : *Solve the system of equations.*

$2x_1 - 4x_2 + 3x_3 - x_4 + x_5 = 2$
$x_1 - 2x_2 + x_4 + 2x_5 = 4$
$4x_1 - 8x_2 + 3x_3 + x_4 + 5x_5 = 10.$

Sol. : First, convert the equation into matrix form $Ax = b$. Form the augmented matrix [A/b] and apply elementary row operations :

$$\left[\begin{array}{ccccc|c} 2 & -4 & 3 & -1 & 1 & 2 \\ 1 & -2 & 0 & 1 & 2 & 4 \\ 4 & -8 & 3 & 1 & 5 & 10 \end{array}\right]$$

$R_1 \leftrightarrow R_2$

$$\sim \left[\begin{array}{ccccc|c} 1 & -2 & 0 & 1 & 2 & 4 \\ 2 & -4 & 3 & -1 & 1 & 2 \\ 4 & -8 & 3 & 1 & 5 & 10 \end{array}\right]$$

$-2R_1 + R_2 \rightarrow R_2$

$$\sim \left[\begin{array}{ccccc|c} 1 & -2 & 0 & 1 & 2 & 4 \\ 0 & 0 & 3 & -3 & -3 & -6 \\ 4 & -8 & 3 & 1 & 5 & 10 \end{array}\right]$$

$R_2/3 \rightarrow R_2$

$$\sim \left[\begin{array}{ccccc|c} 1 & -2 & 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -1 & -1 & -2 \\ 4 & -8 & 3 & 1 & 5 & 10 \end{array}\right]$$

$-4R_1 + R_3 \rightarrow R_3$

$$\sim \left[\begin{array}{ccccc|c} 1 & -2 & 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -1 & -1 & -2 \\ 0 & 0 & 3 & -3 & -3 & -6 \end{array}\right]$$

$-3R_2 + R_3 \rightarrow R_3$

$$\sim \left[\begin{array}{ccccc|c} 1 & -2 & 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The final matrix is in row Echelon form. Since columns 2, 4 and 5 do not contain leading 1's, set x_2, x_4 and x_5 to be free variables. Then using the equation given by the second row,

$$x_3 - x_4 - x_5 = -2 \Rightarrow x_3 = -2 + x_4 + x_5$$

Finally, using the equation given by the first row,

$$x_1 - 2x_2 + x_4 + 2x_5 = 4 \Rightarrow 4 + 2x_2 - x_4 - 2x_5.$$

Summarizing, we can write all solutions of the system in vector form as,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 4 + 2x_2 - x_4 - 2x_5 \\ x_2 \\ -2 + x_4 + x_5 \\ x_4 \\ x_5 \end{bmatrix}, \; x_2, x_4, x_5 \in \mathbb{R}$$

An alternative way to write this solution is,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -2 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Ex. 1.3.4 : *Solve* $x + y + w = 1$, $2x + z + w = 3$, $2y + z + 2w = 2$. **GTU : Winter-22, Marks 7**

Sol. : Convert the augmented matrix into the row Echelon form :

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 1 \\ 2 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 2 & 2 \end{array}\right]$$

$R_2 - 2R_1$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 1 \\ 0 & -2 & 1 & -1 & 1 \\ 0 & 2 & 1 & 2 & 2 \end{array}\right]$$

$R_3 - (-1) R_2$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 1 \\ 0 & -2 & 1 & -1 & 1 \\ 0 & 0 & 2 & 1 & 3 \end{array}\right]$$

$$\left.\begin{array}{r} x+y+w = 1 \\ -2\cdot y+z-w = 1 \\ 2\cdot z+w = 3 \end{array}\right\} \text{... System (1)}$$

Find the variable z from the equation 3 of the system (1) :

$2z = 3 - t$

$z = \frac{3}{2} - \frac{1}{2}\cdot t$

Find the variable y from the equation 2 of the system (1) : $-2y = 1 - z + t = 1 - \left(\frac{3}{2} - \frac{1}{2}\cdot t\right) + t$

$= \frac{-1}{2} + \frac{3}{2}\cdot t$

$y = \frac{1}{4} - \frac{3}{4}\cdot t$

Find the variable x from the equation 1 of the system (1) : $x = 1 - y - t = 1 - \left(\frac{1}{4} - \frac{3}{4}\cdot t\right) - t = \frac{3}{4} - \frac{1}{4}\cdot t$

$$x = \frac{3}{4} - \frac{1}{4}\cdot t$$

$$y = \frac{1}{4} - \frac{3}{4}\cdot t$$

$$z = \frac{3}{2} - \frac{1}{2}\cdot t$$

$$w = t$$

General solution : $X = \begin{pmatrix} \frac{3}{4} - \frac{1}{4}\cdot t \\ \frac{1}{4} - \frac{3}{4}\cdot t \\ \frac{3}{2} - \frac{1}{2}\cdot t \\ t \end{pmatrix}$

Ex. 1.3.5 : *Solve the following equations by Gauss elimination method : $x + y + z = 6$, $x + 2y + 3z = 14$, $2x + 4y + 7z = 30$.* **GTU : Summer-21, Marks 4**

Sol. : Convert the augmented matrix into the row Echelon form :

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 2 & 4 & 7 & 30 \end{array}\right]$$

$R_2 - 1R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 2 & 4 & 7 & 30 \end{array}\right]$$

$R_3 - 2R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 2 & 5 & 18 \end{array}\right]$$

$R_3 - 2R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\left.\begin{array}{r} x+y+z = 6 \\ y+2\cdot z = 8 \\ z = 2 \end{array}\right\} \text{... System (1)}$$

Find the variable z from the equation 3 of the system (1) : $z = 2$.

Find the variable y from the equation 2 of the system (1) : $y = 8 - 2z = 8 - 2 \cdot 2 = 4$

Find the variable x from the equation 1 of the system (1) : $x = 6 - y - z = 6 - 4 - 2 = 0$.

$x = 0$, $y = 4$, $z = 2$

General solution : $X = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}$

Ex. 1.3.6 : *Using Gauss elimination method solve the following system*

$-x + 3y + 4z = 30$

$3x + 2y - z = 9$

$2x - y + 2z = 10.$ **GTU : Winter-20, Marks 4**

Sol. : Convert the augmented matrix into the row Echelon form :

$$\left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ 3 & 2 & -1 & 9 \\ 2 & -1 & 2 & 10 \end{array}\right]$$

$R_2 - (-3) R_1$

$$\sim \left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ 0 & 11 & 11 & 99 \\ 2 & -1 & 2 & 10 \end{array}\right]$$

$R_3 - (-2) R_1$

$$\sim \left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ 0 & 11 & 11 & 99 \\ 0 & 5 & 10 & 70 \end{array}\right]$$

$R_3 - \left(\frac{5}{11}\right) R_2$

$$\sim \left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ 0 & 11 & 11 & 99 \\ 0 & 0 & 5 & 25 \end{array}\right]$$

$$\left.\begin{aligned} -1x + 3y + 4z &= 30 \\ 11y + 11z &= 99 \\ 5z &= 25 \end{aligned}\right\} \quad \text{... System (1)}$$

Find the variable z from the equation 3 of the system (1) : $5z = 25$, $z = 5$.

Find the variable y from the equation 2 of the system (1) : $11y = 99 - 11z = 99 - 11 \cdot 5 = 44$, $y = 4$.

Find the variable x from the equation 1 of the system (1) : $-x = 30 - 3y - 4z = 30 - 3 \cdot 4 - 4 \cdot 5 = -2$, $x = 2$.

$x = 2$, $y = 4$, $z = 5$

General solution : $X = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$

Ex. 1.3.7 : *Solve system of linear equation by Gauss elimination method, if solution exists.*

$x + y + 2z = 9$; $2x + 4y - 3z = 1$; $3x + 6y - 5z = 0$.

GTU : Winter-19, Marks 4

Sol. : Convert the augmented matrix into the row Echelon form :

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right]$$

$R_2 - 2R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 3 & 6 & -5 & 0 \end{array}\right]$$

$R_3 - 3R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right]$$

$R_3 - \left(\frac{3}{2}\right) R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 0 & -1/2 & -3/2 \end{array}\right]$$

$$\left.\begin{aligned} x + y + 2z &= 9 \\ 2y - 7z &= -17 \\ \frac{-1}{2} z &= \frac{-3}{2} \end{aligned}\right\} \quad \text{... System (1)}$$

Find the variable z from the equation 3 of the system (1) : $\frac{-1}{2} z = \frac{-3}{2}$, $z = 3$.

Find the variable y from the equation 2 of the system (1) : $2y = -17 + 7z = -17 + 7 \cdot 3 = 4$, $y = 2$.

Find the variable x from the equation 1 of the system (1) : $x = 9 - y - 2z = 9 - 2 - 2 \cdot 3 = 1$.

$x = 1$, $y = 2$, $z = 3$

General solution : $X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

Solution of system of linear equations by Gauss Jordan method

- A second method of elimination, called Gauss-Jordan elimination after Carl Gauss and Wilhelm Jordan (1842 - 1899), continues the reduction process until a reduced row-Echelon form is obtained. This procedure is demonstrated in the next example.

Ex. 1.3.8 : *Solve the system of linear equations.*

$2x + 4y - 2z = 0$

$3x + 4y = 1$.

Sol. : The augmented matrix of the system of linear equations is,

$$\begin{bmatrix} 2 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{bmatrix}$$

- Using a graphing utility, a computer software program or Gauss-Jordan elimination, you can verify that the reduced row-Echelon form of the matrix is,

$$\begin{bmatrix} 1 & 0 & 5 & 2 \\ 0 & 1 & -3 & -1 \end{bmatrix}.$$

- The corresponding system of equation is,

$x + 5z = 2$

$y - 3z = -1$

- Now, using the parameter t to represent the *nonleading* variable z, you have $x = 2 - 5t$, $y = -1 + 3t$, $z = t$, where t is any real number.
- Note that in this example an arbitrary parameter was assigned to the non leading variable z. You subsequently solved for the leading variables x and y as functions of t.

Ex. 1.3.9 : *Solve the following system of linear equations using the Gauss-Jordan method.*

$x + y + 2z = 8, - x - 2y + 3z = 1, 3x - 7y + 4z = 10.$

GTU : Winter-22, Marks 7

Sol. : Convert the augmented matrix into the row reduced Echelon form :

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right]$$

$R_2 - (-1) R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 3 & -7 & 4 & 10 \end{array}\right]$$

$R_3 - 3R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right]$$

$R_2 / (-1)$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & -10 & -2 & -14 \end{array}\right]$$

$R_3 - (-10) R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & -52 & -104 \end{array}\right]$$

$R_3 / (-52)$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$R_2 - (-5) R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$R_1 - 2R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$R_1 - 1R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\left.\begin{array}{l} x = 3 \\ y = 1 \\ z = 2 \end{array}\right\} \quad \ldots (1)$$

General solution : $X = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$

Ex. 1.3.10 : *Solve following system by using Gauss Jordan method.*

$x + 2y + z - w = -2$

$2x + 3y - z + 2w = 7$

$x + y + 3z - 2w = -6$

$x + y + z + w = 2$

GTU : Winter-21, Marks 7

Sol. : Convert the augmented matrix into the row reduced Echelon form :

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 2 & 3 & -1 & 2 & 7 \\ 1 & 1 & 3 & -2 & -6 \\ 1 & 1 & 1 & 1 & 2 \end{array}\right]$$

$R_2 - 2R_1$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 0 & -1 & -3 & 4 & 11 \\ 1 & 1 & 3 & -2 & -6 \\ 1 & 1 & 1 & 1 & 2 \end{array}\right]$$

$R_3 - 1R_1$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 0 & -1 & -3 & 4 & 11 \\ 0 & -1 & 2 & -1 & -4 \\ 1 & 1 & 1 & 1 & 2 \end{array}\right]$$

$R_4 - 1R_1$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 0 & -1 & -3 & 4 & 11 \\ 0 & -1 & 2 & -1 & -4 \\ 0 & -1 & 0 & 2 & 4 \end{array}\right]$$

$R_2/(-1)$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 0 & 1 & 3 & -4 & -11 \\ 0 & -1 & 2 & -1 & -4 \\ 0 & -1 & 0 & 4 & 4 \end{array}\right]$$

$R_3 - (-1) R_2$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 0 & 1 & 3 & -4 & -11 \\ 0 & 0 & 5 & -5 & -15 \\ 0 & -1 & 0 & 2 & 4 \end{array}\right]$$

$R_4 - (-1) R_2$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 0 & 1 & 3 & -4 & -11 \\ 0 & 0 & 5 & -5 & -15 \\ 0 & 0 & 3 & -2 & -7 \end{array}\right]$$

$R_3/(5)$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 0 & 1 & 3 & -4 & -11 \\ 0 & 0 & 1 & -1 & -3 \\ 0 & 0 & 3 & -2 & -7 \end{array}\right]$$

$R_4 - 3R_3$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 0 & 1 & 3 & -4 & -11 \\ 0 & 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

$R_3 - (-1) R_4$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 0 & 1 & 3 & -4 & -11 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

$R_2 - (-4) R_4$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 1 & -1 & -2 \\ 0 & 1 & 3 & 0 & -3 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

$R_1 - (-1) R_4$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & -3 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

$R_2 - 3R_3$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

$R_1 - 1R_3$

$$\sim\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

$R_1 - 2R_2$

$$\sim\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

Find the variable w from the row 4 : $w = 2$

Find the variable z from the row 3 : $z = -1$

Find the variable y from the row 2 : $y = 0$

Find the variable x from the row 1 : $x = 1$

$x = 1, y = 0, z = -1, w = 2$

Homogeneous system of linear equations :

- Consider systems of linear equations in which each of the constant terms is zero. We call such systems **homogeneous**. For example, a homogeneous system of *m* equations in *n* variables has the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n &= 0 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \ldots + a_{3n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \ldots + a_{mn}x_n &= 0 \end{aligned}$$

- It is easy to see that a homogeneous system must have at least one solution. Specifically, if all variables in a homogeneous system have the value zero, then each of the equations must be satisfied. Such a solution is called **trivial** (or **obvious**). For instance, a homogeneous system of three equations in the three variables x_1, x_2 and x_3 must have $x_1 = 0$, $x_2 = 0$ and $x_3 = 0$ as a trivial solution.

Solving homogeneous system of linear equations

Ex. 1.3.11 : *Solve the system of linear equations.*

$x_1 - x_2 + 3x_3 = 0$

$2x_1 + x_2 + 3x_3 = 0.$

Sol. : Applying Gauss-Jordan elimination to the augmented matrix

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 2 & 1 & 3 & 0 \end{bmatrix}$$

yields the matrix shown below.

$R_2 + (-2) R_1 \rightarrow R_2$

$$\sim \begin{bmatrix} 1 & -1 & 3 & 0 \\ 0 & 3 & -3 & 0 \end{bmatrix}$$

$\left(\frac{1}{3}\right) R_2 \rightarrow R_2$

$$\sim \begin{bmatrix} 1 & -1 & 3 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}$$

$R_1 + R_2 \rightarrow R_1$

$$\sim \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}$$

The system of equations corresponding to this matrix is,

$$x_1 + 2x_3 = 0$$

$$x_2 - x_3 = 0$$

Using the parameter $t = x_3$, the solution set is,

$x_1 = -2t,\ x_2 = t,\ x_3 = t$, t is any real number.

- Example illustrates an important point about homogeneous systems of linear equations. You began with two equations in three variables and discovered that the system has an infinite number of solutions. In general, a homogeneous system with fewer equations than variables has an infinite number of solutions.

Note : Every homogeneous system of linear equations is consistent. Moreover, if the system has fewer equations than variables, then it must have an infinite number of solutions.

Ex. 1.3.12 : *Use Gauss-Jordan algorithm to solve the system of linear equations*

$2x_1 + 2x_2 - x_3 + x_5 = 0$

$-x_1 - x_2 + 2x_3 - 3x_4 + x_5 = 0$

$x_1 + x_2 - 2x_3 - x_5 = 0$

$x_3 + x_4 + x_5 = 0.$ **GTU : Summer-19, Marks 4**

Sol. : Convert the augmented matrix into the row reduced echelon form :

$$\left[\begin{array}{ccccc|c} 2 & 2 & -1 & 0 & 1 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \\ 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right]$$

$R_1/(2)$

$$\sim \left[\begin{array}{ccccc|c} 2 & 1 & -1/2 & 0 & 1/2 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \\ 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right]$$

$R_2 - (-1) R_1$

$$\sim \left[\begin{array}{ccccc|c} 1 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 3/2 & -3 & 3/2 & 0 \\ 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right]$$

$R_3 - 1R_1$

$$\sim \left[\begin{array}{ccccc|c} 1 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 3/2 & -3 & 3/2 & 0 \\ 0 & 0 & -3/2 & 0 & -3/2 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right]$$

$R_2\left(\frac{3}{2}\right)$

$$\sim \left[\begin{array}{ccccc|c} 1 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & -3/2 & 0 & -3/2 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right]$$

$R_3 - \left(\frac{-3}{2}\right) R_2$

$$\sim \left[\begin{array}{ccccc|c} 1 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right]$$

$R_4 - 1R_2$

$$\sim\left[\begin{array}{cccccc|c} 1 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \end{array}\right]$$

$R_3/(-3)$

$$\sim\left[\begin{array}{ccccc|c} 1 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \end{array}\right]$$

$R_4 - 3R_3$

$$\sim\left[\begin{array}{ccccc|c} 1 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$R_2 - (-2) R_3$

$$\sim\left[\begin{array}{ccccc|c} 1 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$R_1 - \left(\frac{-1}{2}\right) R_2$

$$\sim\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{aligned} x_1 + x_2 + x_5 &= 0 \\ x_3 + x_5 &= 0 \\ x_4 &= 0 \end{aligned}\right\} \quad \text{... System (1)}$$

Find the variable x_4 from the equation 3 of the system (1) : $x_4 = 0$.
Find the variable x_3 from the equation 2 of the system (1) : $x_3 = -x_5$.
Find the variable x_1 from the equation 1 of the system (1) : $x_1 = -x_2 - x_5$

$x_1 = -x_2 - x_5,\ x_2 = x_2,\ x_3 = -x_5,\ x_4 = 0,\ x_5 = x_5$

$$\text{General solution : } X = \begin{bmatrix} -x_2 - x_5 \\ x_2 \\ -x_5 \\ 0 \\ x_5 \end{bmatrix}$$

$$\text{The solution set : } \left\{ x_2 \cdot \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \cdot \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

1.4 Rank by Echelon Forms

GTU : Summer-19, Winter-19, 20, 21, 22

- The rank of a matrix A, denoted by Rank (A), is the number of non-zero rows in row Echelon or reduced row echelon form.

Ex. 1.4.1 *Find the rank of the matrix*

$$A = \begin{bmatrix} 1 & 2 & -1 & 0 \\ -1 & 3 & 0 & -4 \\ 2 & 1 & 3 & -2 \\ 1 & 1 & 1 & -1 \end{bmatrix}.$$

GTU : Winter-22, Marks 4

Sol. : The rank of the matrix is equal to the number of non-zero rows in the matrix after reducing it to the row Echelon form using elementary transformations over the rows of the matrix.

$$A = \begin{bmatrix} 1 & 2 & -1 & 0 \\ -1 & 3 & 0 & -4 \\ 2 & 1 & 3 & -2 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

To calculate matrix rank transform matrix to **upper triangular form, using elementary row operations.**

$R_2 + 1R_1 \to R_2$ (Multiply 1 row by 1 and add it to 2 row); $R_3 - 2R_1 \to R_3$ (Multiply 1 row by 2 and subtract it from 3 row); $R_4 - 1R_1 \to R_4$ (Multiply 1 row by 1 and subtract it from 4 row)

$$\sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 5 & -1 & -4 \\ 0 & -3 & 5 & -2 \\ 0 & -1 & 2 & -1 \end{bmatrix}$$

$R_2/5 \rightarrow R_2$ (divide the 2 row by 5)

$$\sim\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & -0.2 & -0.8 \\ 0 & -3 & 5 & -2 \\ 0 & -1 & 2 & -1 \end{bmatrix}$$

$R_3 + 3R_2 \rightarrow R_3$ (Multiply 2 row by 3 and add it to 3 row);

$R_4 + 1R_2 \rightarrow R_4$ (Multiply 2 row by 1 and add it to 4 row)

$$\sim\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & -0.2 & -0.8 \\ \mathbf{0} & \mathbf{0} & \mathbf{4.4} & \mathbf{-4.4} \\ \mathbf{0} & \mathbf{0} & \mathbf{1.8} & \mathbf{-1.8} \end{bmatrix}$$

$R_3/4.4 \rightarrow R_3$ (Divide the 3 row by 4.4)

$$\sim\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & -0.2 & -0.8 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{-1} \\ 0 & 0 & 1.8 & -1.8 \end{bmatrix}$$

$R_4 - 1.8R_3 \rightarrow R_4$ (Multiply 3 row by 1.8 and subtract it from 4 row)

$$\sim\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & -0.2 & -0.8 \\ 0 & 0 & 1 & -1 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

Answer : Since there is 3 **non-zero rows**, then

Rank (A) = 3.

Ex. 1.4.2 : *Find the rank of the matrix* $\begin{bmatrix} 8 & 0 & 0 & 16 \\ 0 & 0 & 0 & 6 \\ 0 & 9 & 9 & 9 \end{bmatrix}$.

GTU : Winter-20, Marks 3

Sol. :

$$A = \begin{bmatrix} 8 & 0 & 0 & 16 \\ 0 & 0 & 0 & 6 \\ 0 & 9 & 9 & 9 \end{bmatrix}$$

To calculate matrix rank transform matrix to **upper triangular form**, using **elementary row operations.**

$R_1/8 \rightarrow R_1$ (Divide the 1 row by 8)

$$\sim\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{2} \\ 0 & 0 & 0 & 6 \\ 0 & 9 & 9 & 9 \end{bmatrix}$$

$R_2 \leftrightarrow R_3$ (Interchange the 2 and 3 rows)

$$\sim\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 9 & 9 & 9 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$R_2/9 \rightarrow R_2$ (Divide the 2 row by 9)

$$\sim\begin{bmatrix} 1 & 0 & 0 & 2 \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$R_3/6 \rightarrow R_3$ (Divide the 3 row by 6)

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}$$

Answer : Since there is 3 **non-zero rows**, then Rank(A) = 3.

Ex. 1.4.3 : *Find the rank of the matrix* $\begin{bmatrix} 0 & 1 & -3 & - \\ 1 & 0 & 1 & \\ 3 & 1 & 0 & \\ 1 & 1 & -2 & \end{bmatrix}$ *by reducing to row Echelon form.*

GTU : Winter-21, Marks

Sol. :

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

To calculate matrix rank transform matrix to upp **triangular form**, using **elementary row operations.**

$R_1 \rightarrow R_2$ (Interchange the 1 and 2 rows)

$$\sim\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

$R_3 - 2R_1 \rightarrow R_3$ (Multiply 1 row by 3 and subtract it from 3 row); $R_4 - 1R_1 \rightarrow R_4$ (Multiply 1 row by 1 and subtract it from 4 row)

$$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{bmatrix}$$

$R_3 - 1R_2 \rightarrow R_3$ (Multiply 2 row by 1 and subtract it from 3 row); $R_4 - 1R_2 \rightarrow R_4$ (Multiply 2 row by 1 and subtract it from 4 row)

$$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Answer : Since there is 2 **non-zero rows,** then Rank(A) = 2.

Ex. 1.4.4 : *Define the rank of a matrix and find the rank of the matrix* $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix}$.

GTU : Summer-19, Marks 3

Sol. :

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix}$$

To calculate matrix rank transform matrix to **upper triangular form,** using **elementary row operations.**

$R_1/2 \rightarrow R_1$ (Divide the 1 row by 2)

$$\begin{bmatrix} 1 & -0.5 & 0 \\ 4 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix}$$

$R_2 - 4R_1 \rightarrow R_2$ (Multiply 1 row by 4 and subtract it from 2 row); $R_3 - 1R_1 \rightarrow R_3$ (Multiply 1 row by 1 and subtract it from 3 row)

$$\sim \begin{bmatrix} 1 & -0.5 & 0 \\ 0 & 7 & -3 \\ 0 & -3.5 & 7 \end{bmatrix}$$

$R_2/7 \rightarrow R_2$ (Divide the 2 row by 7)

$$\sim \begin{bmatrix} 1 & -0.5 & 0 \\ 0 & 1 & -3/7 \\ 0 & -3.5 & 7 \end{bmatrix}$$

$R_3 + 3.5R_2 \rightarrow R_3$ (Multiply 2 row by 3.5 and add it to 3 row)

$$\sim \begin{bmatrix} 1 & -0.5 & 0 \\ 0 & 1 & -3/7 \\ 0 & 0 & 5.5 \end{bmatrix}$$

$R_3/5.5 \rightarrow R_3$ (Divide the 3 row by 5.5)

$$\sim \begin{bmatrix} 1 & -0.5 & 0 \\ 0 & 1 & -3/7 \\ 0 & 0 & 1 \end{bmatrix}$$

Answer : Since there is 3 **non-zero rows,** then Rank(A) = 3.

Ex. 1.4.5 : *Reduce matrix* $A = \begin{bmatrix} 1 & 5 & 3 & -2 \\ 2 & 0 & 4 & 1 \\ 4 & 8 & 9 & -1 \end{bmatrix}$ *to row Echelon form and find its rank.*

GTU : Winter-19, Marks 3

Sol. :

$$A = \begin{bmatrix} 1 & 5 & 9 & -2 \\ 2 & 0 & 4 & 1 \\ 4 & 8 & 9 & -1 \end{bmatrix}$$

To calculate matrix rank transform matrix to **upper triangular form,** using **elementary row operations.**

$R_2 - 2R_1 \rightarrow R_2$ (Multiply 1 row by 2 and subtract it from 2 row); $R_3 - 4R_1 \rightarrow R_3$ (Multiply 1 row by 4 and subtract it from 3 row)

$$\sim \begin{bmatrix} 1 & 5 & 3 & -2 \\ 0 & -10 & -2 & 5 \\ 0 & -12 & -3 & 7 \end{bmatrix}$$

$R_2/-10 \rightarrow R_2$ (Divide the 2 row by – 10)

$$\sim \begin{bmatrix} 1 & 5 & 3 & -2 \\ 0 & 1 & 0.2 & -0.5 \\ 0 & -12 & -3 & 7 \end{bmatrix}$$

$R_3 - 12R_2 \to R_3$ (Multiply 2 row by 12 and add it to 3 row)

$$\sim\begin{bmatrix}1 & 5 & 3 & -2\\0 & 1 & 02 & -0.5\\0 & 0 & -0.6 & 1\end{bmatrix}$$

$R_3/-0.6 \to R_3$ (Divide the 3 row by – 0.6)

$$\sim\begin{bmatrix}1 & 5 & 3 & -2\\0 & 1 & 02 & -0.5\\0 & 0 & 1 & -5/3\end{bmatrix}$$

Answer : Since there is 3 **non-zero rows,** then Rank (A) = 3.

1.5 Inverse of a Square Matrix by Gauss - Jordan Method

GTU : Winter-20, Summer-21, 22, 23

- A way to do this is to augment A by all columns of I to get $[A/I]$ and apply elementary row operations until we obtain $[I/A^{-1}]$:
- This is called the Gauss-Jordan method for finding A^{-1}. Incidentally, even if we do not know whether A is invertible or not when we begin, the row Echelon form of A will reveal this information. Hence we can apply this algorithm without first checking invertibility.

Ex. 1.5.1 : *Find the inverse of the matrix*

$$A = \begin{bmatrix}1 & 2 & 3\\0 & 1 & 4\\5 & 6 & 0\end{bmatrix}.$$

Sol. :

$$[A\,|\,I] = \begin{bmatrix}1 & 2 & 3 & | & 1 & 0 & 0\\0 & 1 & 4 & | & 0 & 1 & 0\\5 & 6 & 0 & | & 0 & 0 & 1\end{bmatrix}$$

$-5R_1 + R_3 \to R_3$

$$\sim\begin{bmatrix}1 & 2 & 3 & | & 1 & 0 & 0\\0 & 1 & 4 & | & 0 & 1 & 0\\0 & -4 & -15 & | & -5 & 0 & 1\end{bmatrix}$$

$4R_2 + R_3 \to R_3$

$$\sim\begin{bmatrix}1 & 2 & 3 & | & 1 & 0 & 0\\0 & 1 & 4 & | & 0 & 1 & 0\\0 & 0 & 1 & | & -5 & 4 & 1\end{bmatrix}$$

$-2R_2 + R_1 \to R_1$

$$\sim\begin{bmatrix}1 & 0 & -5 & | & 1 & -2 & 0\\0 & 1 & 4 & | & 0 & 1 & 0\\0 & 0 & 1 & | & -5 & 4 & 1\end{bmatrix}$$

$5R_3 + R_1 \to R_1$

$$\sim\begin{bmatrix}1 & 0 & 0 & | & -24 & 18 & 5\\0 & 1 & 4 & | & 0 & 1 & 0\\0 & 0 & 1 & | & -5 & 4 & 1\end{bmatrix}$$

$-4R_3 + R_2 \to R_2$

$$\sim\begin{bmatrix}1 & 0 & 0 & | & -24 & 18 & 5\\0 & 1 & 0 & | & 20 & -15 & -4\\0 & 0 & 1 & | & -5 & 4 & 1\end{bmatrix}$$

Therefore A is invertible and

$$A^{-1} = \begin{bmatrix}-24 & 18 & 5\\20 & -15 & -4\\-5 & 4 & 1\end{bmatrix}$$

Ex. 1.5.2 : *Find the inverse of the matrix* $A = \begin{bmatrix}1 & 3 & 3\\1 & 4 & 3\\1 & 3 & 4\end{bmatrix}$ *using Gauss-Jordan method.*

GTU : Summer-22, 23, Marks 4

Sol. : Adjoin the **identity matrix** onto the right of the original matrix, so that you have A on the left side and the identity matrix on the right side. It will look like this :

$$\left[\begin{array}{ccc|ccc}1 & 3 & 3 & 1 & 0 & 0\\1 & 4 & 3 & 0 & 1 & 0\\1 & 3 & 4 & 0 & 0 & 1\end{array}\right]$$

Now find the inverse matrix. Using **elementary row operations** to transform the left side of the resulting matrix to the identity matrix.

$R_2 - 1R_1 \to R_2$ (Multiply 1 row by 1 and subtract it from 2 row); $R_3 - 1R_1 \to R_3$ (Multiply 1 row by 1 and subtract it from 3 row)

$$\sim\left[\begin{array}{ccc|ccc}1 & 3 & 3 & 1 & 0 & 0\\0 & 1 & 0 & -1 & 1 & 0\\0 & 0 & 1 & -1 & 0 & 1\end{array}\right]$$

$R_1 - 3R_2 \rightarrow R_1$(Multiply 2 row by 3 and subtract it from 1 row)

$$\sim\left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 4 & -3 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

$R_1 - 3R_3 \rightarrow R_1$ (Multiply 3 row by 3 and subtract it from 1 row)

$$\sim\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & -3 & -3 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

Answer : $A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

Ex. 1.5.3 : *Use Gauss-Jordan method to find A 1, if* $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$.

GTU : Winter-20, Summer-21, Marks 3

Sol. : Adjoin the **identity matrix** onto the right of the original matrix, so that you have A on the left side and the identity matrix on the right side. It will look like this :

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right]$$

Now find the inverse matrix. Using **elementary row operations** to transform the left side of the resulting matrix to the identity matrix.

$R_2 - 2R_1 \rightarrow R_2$ (Multiply 1 row by 2 and subtract it from 2 row); $R_3 - 1R_1 \rightarrow R_3$ (Multiply 1 row by 1 and subtract it from 3 row)

$$\sim\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right]$$

$R_1 - 2R_2 \rightarrow R_1$ (Multiply 2 row by 2 and subtract it from 1 row); $R_3 - 2R_2 \rightarrow R_3$ (Multiply 2 row by 2 and add it to 3 row)

$$\sim\left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right]$$

$R_3/-1 \rightarrow R_3$ (Divide the 3 row by − 1)

$$\sim\left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$R_1 - 9R_3 \rightarrow R_1$ (Multiply 3 row by 9 and subtract it from 1 row); $R_2 - 3R_3 \rightarrow R_2$ (Multiply 3 row by 3 and add it to 2 row)

$$\sim\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

Answer : $A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$

1.6 Eigenvalues and Eigenvectors

GTU : Winter-19, 20, 21, 22

- **Definition :** Suppose that A is an $n \times n$ matrix. A vector $v \neq 0$ is called an **Eigenvector** for A if $Av = \lambda v$ for some number λ. In this case, λ is called the **Eigenvalue** associated to v.
- How can we find eigenvalues and eigenvectors of a given matrix ? Suppose that $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{Av} = \lambda\mathbf{v}$. Since both λ and **v** are unknown, we cannot solve this linear system directly. But we have

$$\mathbf{Av} - \lambda\mathbf{v} = \mathbf{0} \Rightarrow (\mathbf{A} - \lambda\mathbf{I})\,\mathbf{v} = \mathbf{0}$$

- Since the last equation is a homogeneous system and **v** is a non-trivial solution, the matrix $A - \lambda I$ must be non-invertible. Therefore $\det(A - \lambda I) = 0$:
- **Definition :** The equation $\det(A - \lambda I) = 0$ is called the **Characteristic equation** of the matrix A.

Working rule to find characteristic equation :

For a 3 × 3 matrix :

Method 1 : The characteristic equation is $\det(A - \lambda I) = 0$.

Method 2 : Its characteristic equation can be written as $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$ where,

S_1 = Sum of the main diagonal elements,

S_2 = Sum of the minors of the main diagonal elements,

S_3 = Determinant of A = |A|

For a 2×2 matrix :

Method 1 : The characteristic equation is $|A - \lambda I| = 0$.

Method 2 : Its characteristic equation can be written as $\lambda^2 - S_1\lambda + S_2 = 0$ where S_1 = Sum of the main diagonal elements, S_2 = Determinant of A = $|A|$.

Ex. 1.6.1 : *Find the characteristic equation of the matrix* $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$.

Sol. : Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$. Its characteristic equation is

$\lambda^2 - S_1\lambda + S_2 = 0$ where,

S_1 = Sum of the main diagonal elements = 1 + 2 = 3,

S_2 = Determinant of A = $|A|$ = 1(2) − 2(0) = 2.

Therefore, the characteristic equation is $\lambda^2 - 3\lambda + 2 = 0$.

Ex. 1.6.2 : *Find the characteristic equation of* $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$.

Sol. : Its characteristic equation is

$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$ where

S_1 = Sum of the main diagonal elements

$= 8 + 7 + 3 = 18,$

S_2 = Sum of the minors of the main diagonal elements

$$= \begin{vmatrix} 7 & -4 \\ -4 & 3 \end{vmatrix} + \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}$$

$= 5 + 20 + 20 = 45,$

S_3 = Determinant of A = $|A|$ = 8(5) + 6(− 10) + 2(10)

$= 40 - 60 + 20 = 0.$

Therefore, the characteristic equation is,

$\lambda^3 - 18\lambda^2 + 45\lambda = 0$

Ex. 1.6.3 : *Find characteristic equation of* $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 2 & 0 & 1 \end{bmatrix}$.

GTU : Winter-22, Marks 3

Sol. : From the definition of the eigenvector v corresponding to the eigenvalue λ we have Av = λv.

Then : Av − λv = (A − λI) v = 0

Equation has a non-zero solution if and only if

det(A − λI) = 0

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -1 & 1 \\ 0 & 2-\lambda & 1 \\ 2 & 0 & 1-\lambda \end{vmatrix}$$

$$= -\lambda^3 + 4\lambda^2 - 3\lambda - 4$$

Ex. 1.6.4 : *Find the eigenvalues and the eigenvectors of the matrix* $A = \begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix}$.

Sol. : $\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -2 \\ 3 & -4-\lambda \end{vmatrix} = 0$

$$(1-\lambda)(-4-\lambda)+6 = 0$$

$$\lambda^2 + 3\lambda + 2 = 0$$

$$(\lambda+2)(\lambda+1) = 0$$

Therefore the eigenvalues of A are $\lambda_1 = -2$ and $\lambda_2 = -1$. In order to find the eigenvectors for λ_1, solve the linear system $(A - \lambda_1 I)v = 0$:

$$\left[\begin{array}{cc|c} 3 & -2 & 0 \\ 3 & -2 & 0 \end{array}\right]$$

$-R_1 + R_2 \to R_2$

$$\sim \left[\begin{array}{cc|c} 3 & -2 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$-R_1/3 \to R_1$

$$\sim \left[\begin{array}{cc|c} 1 & -2/3 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

Set v_2 to be a free variable. Then the eigenvectors are $v = \begin{bmatrix} 2v_2/3 \\ v_2 \end{bmatrix}$ where $v_2 \neq 0$ is a real number.

The eigenvectors for λ_2 can be found by solving $(A - \lambda_2 I)v = 0$:

$$\sim \left[\begin{array}{cc|c} 2 & -2 & 0 \\ 3 & -3 & 0 \end{array}\right]$$

$R_1/2 \to R_1$

$$\sim \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 3 & -3 & 0 \end{array}\right]$$

$-3R_1 + R_2 \rightarrow R_2$

$$\sim \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

Again, set v_2 to be free. Then the eigenvectors for $\lambda_2 = -1$ are $v = \begin{bmatrix} v_2 \\ v_2 \end{bmatrix}$ where $v_2 \neq 0$.

Note :

1. Corresponding to n distinct eigenvalues, we get n independent eigenvectors.
2. If 2 or more eigenvalues are equal, it may or may not be possible to get linearly independent eigenvectors corresponding to the repeated eigenvalues.
3. Algebraic multiplicity of an eigenvalue λ is the order of the eigenvalue as a root of the characteristic polynomial (i.e., if λ is a double root, then algebraic multiplicity is 2).
4. Geometric multiplicity of λ is the number of linearly independent eigenvectors corresponding to λ.

Ex. 1.6.5 : *Find the eigenvalues and the eigenvectors of the matrix* $A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$.

GTU : Winter-22, Marks 7

Sol. : From the definition of the eigenvector v corresponding to the eigenvalue λ we have $Av = \lambda v$

Then : $Av - \lambda v = (A - \lambda I) \cdot v = 0$

Equation has a non-zero solution if and only if

$\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -6 & -4 \\ 0 & 4-\lambda & 2 \\ 0 & -6 & -3-\lambda \end{vmatrix} = -\lambda^3 + 2\lambda^2 - \lambda$$

$$= -\lambda \cdot (\lambda^2 - 2\lambda + 1) = -\lambda \cdot (\lambda - 1)^2 = 0$$

1. $\lambda_1 = 0$
2. $\lambda_2 = 1$

For every λ we find its own vectors :

1. $\lambda_1 = 0$

$$A - \lambda_1 I = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A - \lambda I) \cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} 1 & -6 & -4 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & -6 & -3 & 0 \end{array}\right]$$

$R_2/(4)$

$$\sim \left[\begin{array}{ccc|c} 1 & -6 & -4 & 0 \\ \mathbf{0} & \mathbf{1} & \mathbf{1/2} & \mathbf{0} \\ 0 & -6 & -3 & 0 \end{array}\right]$$

$R_3 - (-6)R_2$

$$\left[\begin{array}{ccc|c} 1 & -6 & -4 & 0 \\ 0 & 1 & 1/2 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{array}\right]$$

$R_1 - (-6)R_2$

$$\left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{0} & \mathbf{-1} & \mathbf{0} \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{array}{rl} x_1 - x_3 &= 0 \\ x_2 + 1/2 \cdot x_3 &= 0 \end{array}\right\} \quad \text{... System (1)}$$

Find the variable x_2 from the equation 2 of the system (1) : $x_2 = \frac{-1}{2} \cdot x_3$

Find the variable x_1 from the equation 1 of the system (1) : $x_1 = x_3$.

$$x_1 = x_3$$

$$x_2 = \frac{-1}{2} \cdot x_3$$

$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} x_3 \\ \frac{-1}{2} \cdot x_3 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_3 \cdot \begin{bmatrix} 1 \\ \frac{-1}{2} \\ 1 \end{bmatrix} \right\}$

$$v_1 = \begin{bmatrix} 1 \\ \frac{-1}{2} \\ 1 \end{bmatrix}$$

Note : Algebraic multiplicity of an eigenvalue $\lambda = 0$ is 1 and geometric multiplicity of $\lambda = 0$ is 1,

$$\lambda_2 = 1$$

$$A - \lambda_2 I = \begin{bmatrix} 0 & -6 & -4 \\ 0 & 3 & 2 \\ 0 & -6 & -4 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A - \lambda I)v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} 0 & -6 & -4 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & -6 & -4 & 0 \end{array}\right]$$

$R_1/(-6)$

$$\sim \left[\begin{array}{ccc|c} 0 & 1 & 2/3 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & -6 & -4 & 0 \end{array}\right]$$

$R_2 - 3R_1$

$$\sim \left[\begin{array}{ccc|c} 0 & 1 & 2/3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -6 & -4 & 0 \end{array}\right]$$

$R_3 - (-6)R_1$

$$\left[\begin{array}{ccc|c} 0 & 1 & 2/3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$x_2 + 2/3 \cdot x_3 = 0$ ... System (1)

Find the variable x_2 from the equation 1 of the system (1) : $x_2 = 2/3 \cdot x_3$

$$x_1 = x_1$$
$$x_2 = -2/3 \cdot x_3$$
$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} x_1 \\ \frac{-2}{3} \cdot x_3 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_1 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_3 \cdot \begin{bmatrix} 0 \\ \frac{-2}{3} \\ 1 \end{bmatrix} \right\}$

$$v_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} 0 \\ \frac{-2}{3} \\ 1 \end{bmatrix}$$

Note : Algebraic multiplicity of an eigenvalue $\lambda = 1$ is 2 and geometric multiplicity of $\lambda = 1$ is 2.

Ex. 1.6.6 : *Find the eigenvalues and corresponding eigenvectors of the matrix*

$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

GTU : Winter-21, Marks 7

Sol. : From the definition of the eigenvector v corresponding to the eigenvalue λ we have $Av = \lambda v$.

Then : $Av - \lambda v = (A - \lambda I) \cdot v = 0$

Equation has a non-zero solution if and only if $\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 0 & 1 \\ -2 & 1-\lambda & 0 \\ -2 & 0 & 1-\lambda \end{vmatrix}$$

$$= -\lambda^3 + 6\lambda^2 - 11\lambda + 6 = -(\lambda - 1) \cdot (\lambda^2 - 5\lambda + 6)$$

$$= -(\lambda - 1) \cdot (\lambda - 2) \cdot (\lambda - 3) = 0$$

1. $\lambda_1 = 1$
2. $\lambda_2 = 2$
3. $\lambda_3 = 3$

For every λ we find its own vectors :

1. $\lambda_1 = 1$

$$A-\lambda_1 I = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A-\lambda I)v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} 3 & 0 & 1 & 0 \\ -2 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 \end{array}\right]$$

$R_1/(3)$

$$\sim \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{0} & \mathbf{1/3} & \mathbf{0} \\ -2 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 \end{array}\right]$$

$R_2 -(-2)R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1/3 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{2/3} & \mathbf{0} \\ -2 & 0 & 0 & 0 \end{array}\right]$$

$R_3 -(-2)R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1/3 & 0 \\ 0 & 0 & 2/3 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{2/3} & \mathbf{0} \end{array}\right]$$

$R_2/[2/3]$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1/3 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ 0 & 0 & 2/3 & 0 \end{array}\right]$$

$R_3 -(2/3)R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1/3 & 0 \\ 0 & 0 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{array}\right]$$

$R_1 -[1/3]R_2$

$$\sim \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

General solution : $X = \begin{bmatrix} 0 \\ x_2 \\ 0 \end{bmatrix}$

The solution set : $\left\{ x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

$$v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda_2 = 2$$

$$A-\lambda_2 I = \begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A-\lambda I)v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} 2 & 0 & 1 & 0 \\ -2 & -1 & 0 & 0 \\ -2 & 0 & -1 & 0 \end{array}\right]$$

$R_1/(2)$

$$\sim \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{0} & \mathbf{1/2} & \mathbf{0} \\ -2 & -1 & 0 & 0 \\ -2 & 0 & -1 & 0 \end{array}\right]$$

$R_2 -(-2)\cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1/2 & 0 \\ 0 & -1 & 1 & 0 \\ -2 & 0 & -1 & 0 \end{array}\right]$$

$R_3 -(-2)\cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1/2 & 0 \\ 0 & -1 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{array}\right]$$

$R_2/(-1)$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1/2 & 0 \\ \mathbf{0} & \mathbf{1} & \mathbf{-1} & \mathbf{0} \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{aligned} x_1 + \frac{1}{2}\cdot x_3 &= 0 \\ x_2 - x_3 &= 0 \end{aligned}\right\} \quad \text{... System (1)}$$

Find the variable x_2 from the equation 2 of the system (1) : $x_2 = x_3$.
Find the variable x_1 from the equation 1 of the system (1) : $x_1 = \frac{-1}{2}x_3$.

$$x_1 = \frac{-1}{2}\cdot x_3$$

$$x_2 = x_3$$

$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} \frac{-1}{2}\cdot x_3 \\ x_3 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_3 \cdot \begin{bmatrix} \frac{-1}{2} \\ 1 \\ 1 \end{bmatrix} \right\}$

$$v_2 = \begin{bmatrix} \frac{-1}{2} \\ 1 \\ 1 \end{bmatrix}$$

$$\lambda_3 = 3$$

$$A - \lambda_3 I = \begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A - \lambda I)\cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ -2 & -2 & 0 & 0 \\ -2 & 0 & -2 & 0 \end{array}\right]$$

$R_2 - (-2)\cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & -2 & 2 & 0 \\ -2 & 0 & -2 & 0 \end{array}\right]$$

$R_3 - (-2)\cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$R_2/(-2)$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{aligned} x_1 + x_3 &= 0 \\ x_2 - x_3 &= 0 \end{aligned}\right\} \quad \text{... System (1)}$$

Find the variable x_2 from the equation 2 of the system (1) : $x_2 = x_3$.
Find the variable x_1 from the equation 1 of the system (1) : $x_1 = x_3$.

$$x_1 = -x_3$$
$$x_2 = x_3$$
$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} -x_3 \\ x_3 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$v_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

Ex. 1.6.7 : *If 1 is an eigenvalue of the matrix* $\begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ *then find is corresponding eigenvector.*

GTU : Winter-21, Marks 3

Sol. :

$$\lambda_3 = 1$$

$$A - \lambda_3 I = \begin{bmatrix} 1 & 1 & 0 \\ -1 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A-\lambda I)\cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$R_2 - (-1)R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{array}{r} x_1 + x_2 = 0 \\ x_3 = 0 \end{array}\right\} \quad \text{... System (1)}$$

Find the variable x_3 from the equation 2 of the system (1) : $x_3 = 0$.

Find the variable x_1 from the equation 1 of the system (1) : $x_1 = -x_2$.

$$x_1 = -x_2$$

$$x_2 = x_2$$

$$x_3 = 0$$

General solution : $X = \begin{bmatrix} -x_2 \\ x_2 \\ 0 \end{bmatrix}$

The solution set : $\left\{ x_2 \cdot \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$

$$v_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Ex. 1.6.8 : *Find the eigenvalues and eigenvectors for the matrix* $\begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$.

GTU : Winter-20, Marks 4

Sol. : From the definition of the eigenvector v corresponding to the eigenvalue λ we have $Av = \lambda v$

Then : $Av - \lambda v (A - \lambda I)\cdot v = 0$

Equation has a non-zero solution if and only if

$\det(A - \lambda I) = 0$

$$\det(A-\lambda I) = \begin{vmatrix} 0-\lambda & 0 & -2 \\ 1 & 2-\lambda & 1 \\ 1 & 0 & 3-\lambda \end{vmatrix} = -\lambda^3 + 5\lambda^2 - 8\lambda + 4$$

$$= -(\lambda-1)(\lambda^2 - 4\lambda + 4) = -(\lambda-1)(\lambda-2)^2 = 0$$

1. $\lambda_1 = 1$
2. $\lambda_2 = 2$

For every λ we find its own vectors :

1. $\lambda_1 = 1$

$$A - \lambda_1 I = \begin{bmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A-\lambda I)\cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} -1 & 0 & -2 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right]$$

$R_1/(-1)$

$$\sim \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{0} & \mathbf{2} & \mathbf{0} \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right]$$

$R_2 - 1\cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ \mathbf{0} & \mathbf{1} & \mathbf{-1} & \mathbf{0} \\ 1 & 0 & 2 & 0 \end{array}\right]$$

$R_3 - 1\cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{array}\right]$$

$$\left.\begin{array}{r} x_1 + 2\cdot x_3 = 0 \\ x_2 - x_3 = 0 \end{array}\right\} \quad \text{... System (1)}$$

Find the variable x_2 from the equation 2 of the system (1) : $x_2 = x_3$.

Find the variable x_1 from the equation 1 of the system (1) : $x_1 = -2x_3$.

$$x_1 = -2x_3$$
$$x_2 = x_3$$
$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} -2x_3 \\ x_3 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_3 \cdot \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$v_1 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 2$$

$$A - \lambda_2 I = \begin{bmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A - \lambda I) \cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} -2 & 0 & -2 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{array}\right]$$

$R_1/(-2)$

$$\sim \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{array}\right]$$

$R_2 - 1 \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 1 & 0 & 1 & 0 \end{array}\right]$$

$R_3 - 1 \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{array}\right]$$

$x_1 + x_3 = 0$ ... System (1)

Find the variable x_1 from the equation 1 of the system (1) : $x_1 = -x_3$.

$$x_1 = -x_3$$
$$x_2 = x_2$$
$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} -x_3 \\ x_2 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_2 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \cdot \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Ex. 1.6.9 : *Find the eigenvalues and eigenvectors for the matrix* $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}$.

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Sol. : From the definition of the eigenvector v corresponding to the eigenvalue λ we have $Av = \lambda v$

Then : $Av - \lambda v (A - \lambda I) \cdot v = 0$

Equation has a non-zero solution if and only if

$\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = \begin{vmatrix} 0-\lambda & 1 & 0 \\ 0 & 0-\lambda & 1 \\ 1 & -3 & 3-\lambda \end{vmatrix}$$

$$= -\lambda^3 + 3\lambda^2 - 3\lambda + 1$$

$$= -(\lambda - 1)^3 = \lambda_1 = 1$$

For every λ we find its own vectors :

1. $\lambda_1 = 1$

$$A - \lambda_1 I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & -3 & 2 \end{bmatrix}$$

$$\therefore \quad Av = \lambda v^*$$

$$(A - \lambda I) \cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & -3 & 2 & 0 \end{array}\right]$$

$R_1/(-1)$

$$\sim \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{-1} & \mathbf{0} & \mathbf{0} \\ 0 & -1 & 1 & 0 \\ 1 & -3 & 2 & 0 \end{array}\right]$$

$R_3 - 1 \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ \mathbf{0} & \mathbf{-2} & \mathbf{2} & \mathbf{0} \end{array}\right]$$

$R_2/(-1)$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ \mathbf{0} & \mathbf{1} & \mathbf{-1} & \mathbf{0} \\ 0 & -2 & 2 & 0 \end{array}\right]$$

$R_3 - (-2) \cdot R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{array}\right]$$

$R_1 - (-1) \cdot R_2$

$$\sim \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{0} & \mathbf{-1} & \mathbf{0} \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{array}{l} x_1 - x_3 = 0 \\ x_2 - x_3 = 0 \end{array}\right\} \quad \text{... System (1)}$$

Find the variable x_2 from the equation 2 of the system (1) : $x_2 = x_3$.

Find the variable x_1 from the equation 1 of the system (1) : $x_1 = x_3$.

$$x_1 = x_3$$
$$x_2 = x_3$$
$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} x_3 \\ x_3 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Note : Algebraic multiplicity of an eigenvalue $\lambda = 1$ is 3 and geometric multiplicity of $\lambda = 1$ is 1.

Ex. 1.6.10 : *Find the eigenvalues and eigenvectors for the matrix A where* $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$.

Sol. : From the definition of the eigenvector v corresponding to the eigenvalue λ we have $Av = \lambda v$.

Then : $Av - \lambda v (A - \lambda I) \cdot v = 0$

Equation has a non-zero solution if and only if

$\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 2 & 2 \\ 0 & 2-\lambda & 1 \\ -1 & 2 & 2-\lambda \end{vmatrix}$$

$= -\lambda^3 + 5\lambda^2 - 8\lambda + 4 = -(\lambda - 1)(\lambda^2 - 4\lambda + 4)$

$= -(\lambda - 1) \cdot (\lambda - 2)^2 = 0$

$\lambda_1 = 1$

$\lambda_2 = 2$

For every λ we find its own vectors :

1. $\lambda_1 = 1$

$$A - \lambda_1 I = \begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A - \lambda I) \cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} 0 & 2 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 2 & 1 & 0 \end{array}\right]$$

R_3

$$\sim\left[\begin{array}{ccc|c} -1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 2 & 0 \end{array}\right]$$

$R_1/(-1)$

$$\sim\left[\begin{array}{ccc|c} 1 & -2 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 2 & 0 \end{array}\right]$$

$R_3 - 2 \cdot R_2$

$$\sim\left[\begin{array}{ccc|c} 1 & -2 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$R_1 - (-2) \cdot R_2$

$$\sim\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{array}{l} x_1 + x_3 = 0 \\ x_2 + x_3 = 0 \end{array}\right\} \quad \text{... System (1)}$$

Find the variable x_2 from the equation 2 of the system (1) : $x_2 = -x_3$.
Find the variable x_1 from the equation 1 of the *system (1)* : $x_1 = -x_3$.

$$x_1 = -x_3$$
$$x_2 = -x_3$$
$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} -x_3 \\ -x_3 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_3 \cdot \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}$

$$v_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 2$$

$$A - \lambda_2 I = \begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A - \lambda I) \cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} -1 & 2 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 2 & 0 & 0 \end{array}\right]$$

$R_1/(-1)$

$$\sim\left[\begin{array}{ccc|c} 1 & -2 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 2 & 0 & 0 \end{array}\right]$$

$R_3 - (-1) \cdot R_1$

$$\sim\left[\begin{array}{ccc|c} 1 & -2 & -2 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & -2 & 0 \end{array}\right]$$

$R_3 - (-2) \cdot R_2$

$$\sim\left[\begin{array}{ccc|c} 1 & -2 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$R_1 - (-2) \cdot R_2$

$$\sim\left[\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{array}{r} x_1 - 2 \cdot x_2 = 0 \\ x_3 = 0 \end{array}\right\} \quad \text{... System (1)}$$

Find the variable x_3 from the equation 2 of the system (1) : $x_3 = 0$.
Find the variable x_1 from the equation 1 of the system (1) : $x_1 = 2x_2$.

$$x_1 = 2x_2$$
$$x_2 = x_2$$
$$x_3 = 0$$

General solution : $X = \begin{bmatrix} 2x_2 \\ x_2 \\ 0 \end{bmatrix}$

The solution set : $\left\{ x_2 \cdot \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$

$$v_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Note : Algebraic multiplicity of an eigenvalue $\lambda = 2$ is 2 and geometric multiplicity of $\lambda = 1$ is 1.

Alternative method for finding eigenvectors :

Ex. 1.6.11 : *Find the eigenvalues and eigenvectors of* $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$.

Sol. : Let $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ which is a non-symmetric matrix.

To find the characteristic equation :

Its characteristic equation can be written as

$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$ where

S_1 = Sum of the main diagonal elements

$= 2 + 3 + 2 = 7,$

S_2 = Sum of the minors of the main diagonal elements

$= \begin{vmatrix} 3 & 1 \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 1 & 3 \end{vmatrix}$

$= 4 + 3 + 4 = 11,$

S_3 = Determinant of $A = |A| = 2(4) - 2(1) + 1(-1) = 5.$

Therefore, the characteristic equation of A is

$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$

$$\begin{array}{c|cccc} 1 & 1 & -7 & 11 & -5 \\ & 0 & 1 & -6 & 5 \\ \hline & 1 & -6 & 5 & 0 \end{array}$$

$(\lambda - 1)(\lambda^2 - 6\lambda + 5) = 0 \Rightarrow \lambda = 1,$

$$\lambda = \frac{6 \pm \sqrt{(-6)^2 - 4(1)(5)}}{2(1)} = \frac{6 \pm \sqrt{16}}{2} = \frac{6 \pm 4}{2} = \frac{6+4}{2},$$

$\frac{6-4}{2} = 5, 1$

Therefore, the eigenvalues are 1, 1 and 5

A is non-symmetric matrix with repeated eigenvalues.

To find the eigenvectors :

$$[A - \lambda I]X = 0$$

$$\begin{bmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Case 1 : If $\lambda = 5$, $\begin{bmatrix} 2-5 & 2 & 1 \\ 1 & 3-5 & 1 \\ 1 & 2 & 2-5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

i.e., $\begin{bmatrix} -3 & 2 & 1 \\ 1 & -2 & 1 \\ 1 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow -3x_1 + 2x_2 + x_3 = 0$ ...(1)

$x_1 - 2x_2 + x_3 = 0$ ...(2)

$x_1 + 2x_2 - 3x_3 = 0$ ...(3)

Considering equations (1) and (2) and using method of cross-multiplication, we get, $x_1 x_2 x_3$.

$$\begin{array}{cccc} 2 & 1 & -3 & 2 \\ -2 & 1 & 1 & -2 \end{array}$$

$$\Rightarrow \quad \frac{x_1}{4} = \frac{x_2}{4} = \frac{x_3}{4} \Rightarrow \frac{x_1}{1} = \frac{x_2}{1} = \frac{x_3}{1}$$

Therefore, $X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Case 2 : If $\lambda = 1$, $\begin{bmatrix} 2-1 & 2 & 1 \\ 1 & 3-1 & 1 \\ 1 & 2 & 2-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

i.e. $\begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Rightarrow x_1 + 2x_2 + x_3 = 0$$

$$x_1 + 2x_2 + x_3 = 0$$

$$x_1 + 2x_2 + x_3 = 0$$

All the three equations are one and the same.

Therefore, $x_1 + 2x_2 + x_3 = 0$

Put $x_1 = 0 \Rightarrow 2x_2 + x_3 = 0 \Rightarrow 2x_2 = -x_3$

Taking $x_3 = 2$, $x_2 = -1$

Therefore, $X_2 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$

Put $x_2 = 0 \Rightarrow x_1 + x_3 = 0 \Rightarrow x_3 = -x_1$

Taking $x_1 = 1$, $x_3 = -1$

Therefore, $X_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Ex. 1.6.12 : *Find the eigenvalues and eigenvectors of* $\begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$.

Sol. : *Let* $A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ which is a non-symmetric matrix.

To find the characteristic equation :

Its characteristic equation can be written as

$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$ where

$S_1 =$ Sum of the main diagonal elements

$= 2 + 1 - 1 = 2,$

$S_2 =$ Sum of the minors of the main diagonal elements

$$= \begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 1 & -1 \end{vmatrix} + \begin{vmatrix} 2 & -2 \\ 1 & 1 \end{vmatrix}$$

$= -4 - 4 + 4 = -4,$

$S_3 =$ Determinant of $A = |A|$

$= 2(-4) + 2(-2) + 2(2) = -8 - 4 + 4 = -8.$

Therefore, the characteristic equation of A is

$\lambda^3 - 2\lambda^2 - 4\lambda + 8 = 0$

$$\begin{array}{c|cccc} 2 & 1 & -2 & -4 & 8 \\ & 0 & 2 & 0 & -8 \\ \hline & 1 & 0 & -4 & \boxed{0} \end{array}$$

$(\lambda - 2)(\lambda^2 - 4) = 0 \Rightarrow \lambda = 1, \lambda = 2, -2$

Therefore, the eigenvalues are 2, 2 and -2

A is non-symmetric matrix with repeated eigenvalues

To find the eigenvectors :

$$[A - \lambda I]X = 0$$

$$\begin{bmatrix} 2-\lambda & -2 & 2 \\ 1 & 1-\lambda & 1 \\ 1 & 3 & -1-\lambda \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Case 1 : If $\lambda = -2$,

$$\begin{bmatrix} 2-(-2) & -2 & 2 \\ 1 & 1-(-2) & 1 \\ 1 & 3 & -1-(-2) \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

i.e., $\begin{bmatrix} 4 & -2 & 2 \\ 1 & 3 & 1 \\ 1 & 3 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow 4x_1 - 2x_2 + 2x_3 = 0$ ...(1)

$x_1 + 3x_2 + x_3 = 0$ ...(2)

$x_1 + 3x_2 + x_3 = 0$ ...(3)

Equations (2) and (3) are one and the same.

Considering equations (1) and (2) and using method of cross-multiplication, we get, $x_1 x_2 x_3$

$$\begin{array}{cccc} -1 & 1 & 2 & -1 \\ 3 & 1 & 1 & 3 \end{array}$$

$$\Rightarrow \quad \frac{x_1}{-4} = \frac{x_2}{-1} = \frac{x_3}{7} \Rightarrow \frac{x_1}{4} = \frac{x_2}{1} = \frac{x_3}{-7}$$

Therefore, $X_1 = \begin{bmatrix} 4 \\ 1 \\ -7 \end{bmatrix}$

Case 2 : If $\lambda = 2$, $\begin{bmatrix} 2-2 & -2 & 2 \\ 1 & 1-2 & 1 \\ 1 & 3 & -1-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

i.e., $\begin{bmatrix} 0 & -2 & 2 \\ 1 & -1 & 1 \\ 1 & 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Rightarrow 0x_1 - 2x_2 + 2x_3 = 0 \qquad \ldots(1)$$

$$x_1 - x_2 + x_3 = 0 \qquad \ldots(2)$$

$$x_1 + 3x_2 - 3x_3 = 0 \qquad \ldots(3)$$

Considering equations (1) and (2) and using method of cross-multiplication, we get, $x_1 x_2 x_3$

$$\begin{matrix} -2 & & 2 & & 0 & & -2 \\ -1 & & 1 & & 1 & & -1 \end{matrix}$$

$$\Rightarrow \quad \frac{x_1}{0} = \frac{x_2}{2} = \frac{x_3}{2} \Rightarrow \frac{x_1}{0} = \frac{x_2}{1} = \frac{x_3}{1}$$

Therefore, $X_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

We get one eigenvector corresponding to the repeated $\lambda_2 = \lambda_3 = 2$.

Problems under properties of eigenvalues and eigenvectors :

Ex. 1.6.13 : *Find the sum and product of the eigenvalues of the matrix* $\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$.

Sol. : Sum of the eigenvalues = Sum of the main diagonal elements = – 3.

Product of the eigenvalues = $|A|$

$= -1(1-1) - (-1-1) + 1(1-(-1)) = 2 + 2 = 4.$

Ex. 1.6.14 : *If 2, 2, 3 are the eigenvalues of* $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$, *find the eigenvalues of* A^T.

Sol. : By the property "A square matrix A and its transpose A^T have the same eigenvalues", the eigenvalues of A^T are 2, 2, 3.

Ex. 1.6.15 : *Find the eigenvalues of* $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 4 & 4 \end{bmatrix}$.

Sol. : Given $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 4 & 4 \end{bmatrix}$. Clearly, A is a lower triangular matrix. Hence, by the property "the characteristic roots of a triangular matrix are just the diagonal elements of the matrix", the eigenvalues of A are 2, 3, 4.

Ex. 1.6.16 : *Find the eigenvalues of the inverse of the matrix* $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 4 \end{bmatrix}$.

Sol. : We know that A is an upper triangular matrix. Therefore, the eigenvalues of A are 2, 3, 4. Hence, by using the property "If the eigenvalues of A are λ_1, λ_2, λ_3, then the eigenvalues of A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3}$, the eigenvalues of A^{-1} are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$.

Ex. 1.6.17 : *Find the eigenvalues of* A^3 *given* $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -7 \\ 0 & 0 & 3 \end{bmatrix}$.

Sol. : Given $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -7 \\ 0 & 0 & 3 \end{bmatrix}$. A is an upper triangular matrix. Hence, the eigenvalues of A are 1, 2, 3.

Therefore, the eigenvalues of A^3 are $1^3, 2^3, 3^3$ i.e., 1, 8, 27.

Ex. 1.6.18 : *Find the eigenvalues of A,* A^2, A^3, A^4, $3A$, A^{-1}, $A - I$ *if* $A = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix}$.

Sol. : Given $A = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix}$. A is an upper triangular matrix. Hence, the eigenvalues of A are 2, 5.

The eigenvalues of A^2 are $2^2, 5^2$ i.e. 4, 25.

The eigenvalues of A^3 are $2^3, 5^3$ i.e. 8, 125.

The eigenvalues of A^4 are $2^4, 5^4$ i.e. 16, 625.

The eigenvalues of 3A are 3(2), 3(5) i.e. 6, 15.

The eigenvalues of A^{-1} are $\frac{1}{2}, \frac{1}{5}$

$$A - 1 = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 4 \end{bmatrix}$$

Since A – I is an upper triangular matrix, the eigenvalues of A – I are its main diagonal elements i.e. 1, 4.

Note : In above examples, eigenvectors remain same as of A.

Non-symmetric matrix : If a square matrix A is non-symmetric, then $A \neq A^T$.

Note :

1. In a non-symmetric n × n matrix, if the eigenvalues are non-repeated then we get **n** linearly independent set of eigenvectors.
2. In a non-symmetric matrix, if the eigenvalues are repeated, then it may or may not be possible to get **n** linearly independent eigenvectors.

   If we form a linearly independent set of **n** eigenvectors, then diagonalization is possible.

Symmetric matrix :

If a square matrix A is symmetric, then $A = A^T$.

Note : In a symmetric matrix, diagonalization is possible.

1.7 Diagonalization of A Matrix

GTU : Summer-19, Winter-22

- By diagonalization, we mean for a given matrix, we shall find a diagonal matrix which is similar to it. Of course, this is not always possible.

Note : An n × n matrix A is diagonalizable, if and only if A has n linearly independent eigenvectors.

- In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A. In this case, the diagonal entries of D are eigenvalues of A that corresponding, respectively, to the eigenvectors in P.

Ex. 1.7.1 : *Diagonalize the following matrix, if possible*

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}.$$

Sol. : Step 1 : Find the eigenvalues of A. In order to find the eigenvalues, we need to set up the characteristic equation and to solve it. In this problem, we have

$$\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 3 & 3 \\ -3 & -5-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{bmatrix}$$

$$= -\lambda^3 - 3\lambda^2 + 4$$

$$= -(\lambda-1)(\lambda+2)^2$$

Therefore, the eigenvalues of A are $\lambda = 1$ and $\lambda = -2$

Step 2 : Find three linearly independent eigenvectors of A. If it is impossible to find three linearly independent eigenvectors, then the matrix is not diagonalizable.

For $\lambda = 1$, we consider the matrix

$$A - 1 \cdot I = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix}$$

In order to solve the homogeneous system $(A - I)x = 0$, we use

$$\begin{bmatrix} 0 & 3 & 3 & 0 \\ -3 & -6 & -3 & 0 \\ 3 & 3 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Thus the homogeneous system becomes

$$x_1 - x_3 = 0, \quad x_2 + x_3 = 0$$

The solution can be written as

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ -x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Thus

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

is an eigenvector.

For $\lambda = -2$, we consider the homogeneous system

$(A + 2I)x = 0$.

$$A + 2I = \begin{bmatrix} 3 & 3 & 3 & 0 \\ -3 & -3 & -3 & 0 \\ 3 & 3 & 3 & 0 \end{bmatrix}$$

Thus we have

$$\begin{bmatrix} 3 & 3 & 3 & 0 \\ -3 & -3 & -3 & 0 \\ 3 & 3 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The system becomes

$$x_1 + x_2 + x_3 = 0$$

and

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_2 - x_3 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \; v_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

are the two other eigenvalues so that v_1, v_2, v_3 are linearly independent.

Step 3 : Construct P from the eigenvectors from step 2. We have,

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Step 4 : Construct D from the corresponding eigenvalues.

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Note : The order of the eigenvectors is not important. For example, we can take

$$P = \begin{bmatrix} -1 & 1 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

and just need to change D to be

$$D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

Note : An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Ex. 1.7.2 : *Show that* $A = \begin{bmatrix} 0 & 1 & 0 \\ -2 & -1 & 2 \\ -4 & -8 & 7 \end{bmatrix}$ *is diagonalizable.*

Find the matrix of eigenvectors and diagonal matrix.

GTU : Winter-22, Marks 7

Sol. : From the definition of the eigenvector v corresponding to the eigenvalue λ we have $Av = \lambda v$.

Then : $Av - \lambda v (A - \lambda I) \cdot v = 0$

Equation has a non-zero solution if and only if

$\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = \begin{vmatrix} 0-\lambda & 1 & 0 \\ -2 & -1-\lambda & 2 \\ -4 & -8 & 7-\lambda \end{vmatrix}$$

$$= -\lambda^3 + 6\lambda^2 - 11\lambda + 6 = -(\lambda - 1) \cdot (\lambda^2 - 5\lambda + 6)$$

$$= -(\lambda - 1) \cdot (\lambda - 2) \cdot (\lambda - 3) = 0$$

1. $\lambda_1 = 1$
2. $\lambda_2 = 2$
3. $\lambda_3 = 3$

For every λ we find its own vectors :

1. $\lambda_1 = 1$

$$A-\lambda_1 I = \begin{bmatrix} -1 & 1 & 0 \\ -2 & -2 & 2 \\ -4 & -8 & 6 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A-\lambda I)\cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} -1 & 1 & 0 & 0 \\ -2 & -2 & 2 & 0 \\ -4 & -8 & 6 & 0 \end{array}\right]$$

$R_1/(-1)$

$$\sim\left[\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ -2 & -2 & 2 & 0 \\ -4 & -8 & 6 & 0 \end{array}\right]$$

$R_2 -(-2)\cdot R_1$

$$\sim\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & -4 & 2 & 0 \\ -4 & -8 & 6 & 0 \end{array}\right]$$

$R_3 -(-4)\cdot R_1$

$$\sim\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & -4 & 2 & 0 \\ 0 & -12 & 6 & 0 \end{array}\right]$$

$R_2/(-4)$

$$\sim\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & -12 & 6 & 0 \end{array}\right]$$

$R_3 -(-12)\cdot R_2$

$$\sim\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$R_1 -(-1)\cdot R_2$

$$\sim\left[\begin{array}{ccc|c} 1 & 0 & -1/2 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{aligned} x_1 - \frac{1}{2}\cdot x_3 &= 0 \\ x_2 - \frac{1}{2}\cdot x_3 &= 0 \end{aligned}\right\} \quad \text{... System (1)}$$

Find the variable x_2 from the equation 2 of the system (1) : $x_2 = \frac{1}{2}\cdot x_3$.

Find the variable x_1 from the equation 1 of the system (1) : $x_1 = \frac{1}{2}\cdot x_3$.

$$x_1 = \frac{1}{2}\cdot x_3$$

$$x_2 = \frac{1}{2}\cdot x_3$$

$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} \frac{1}{2}\cdot x_3 \\ \frac{1}{2}\cdot x_3 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_3 \cdot \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix} \right\}$

$$v_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix}$$

$$\lambda_2 = 2$$

$$A-\lambda_2 I = \begin{bmatrix} -2 & 1 & 0 \\ -2 & -3 & 2 \\ -4 & -8 & 5 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A-\lambda I)\cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} -2 & 1 & 0 & 0 \\ -2 & -3 & 2 & 0 \\ -4 & -8 & 5 & 0 \end{array}\right]$$

$R_1/(-2)$

$$\sim\left[\begin{array}{ccc|c} 1 & -1/2 & 0 & 0 \\ -2 & -3 & 2 & 0 \\ -4 & -8 & 5 & 0 \end{array}\right]$$

$R_2 - (-2) \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & -1/2 & 0 & 0 \\ 0 & -4 & 2 & 0 \\ -4 & -8 & 5 & 0 \end{array}\right]$$

$R_3 - (-4) \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & -1/2 & 0 & 0 \\ 0 & -4 & 2 & 0 \\ 0 & -10 & 5 & 0 \end{array}\right]$$

$R_2/(-4)$

$$\sim \left[\begin{array}{ccc|c} 1 & -1/2 & 0 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & -10 & 5 & 0 \end{array}\right]$$

$R_3 - (-10) \cdot R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & -1/2 & 0 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$R_1 - \left(\frac{-1}{2}\right) \cdot R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -1/4 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{array}{l} x_1 - \frac{1}{4} \cdot x_3 = 0 \\ x_2 - \frac{1}{2} \cdot x_3 = 0 \end{array}\right\} \quad \text{... System (1)}$$

Find the variable x_2 from the equation 2 of the system (1) : $x_2 = \frac{1}{2} \cdot x_3$.

Find the variable x_1 from the equation 1 of the system (1) : $x_1 = \frac{1}{4} \cdot x_3$.

$$x_1 = \frac{1}{4} \cdot x_3$$

$$x_2 = \frac{1}{2} \cdot x_3$$

$$x_3 = x_3$$

$$\text{General solution : } X = \begin{bmatrix} \frac{1}{4} \cdot x_3 \\ \frac{1}{2} \cdot x_3 \\ x_3 \end{bmatrix}$$

$$\text{The solution set : } \left\{ x_3 \cdot \begin{bmatrix} \frac{1}{4} \\ \frac{1}{2} \\ 1 \end{bmatrix} \right\}$$

$$v_2 = \begin{bmatrix} \frac{1}{4} \\ \frac{1}{2} \\ 1 \end{bmatrix}$$

$$\lambda_3 = 3$$

$$A - \lambda_3 I = \begin{bmatrix} -3 & 1 & 0 \\ -2 & -4 & 2 \\ -4 & -8 & 4 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A - \lambda I) \cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} -3 & 1 & 0 & 0 \\ -2 & -4 & 2 & 0 \\ -4 & -8 & 4 & 0 \end{array}\right]$$

$R_1/(-3)$

$$\sim \left[\begin{array}{ccc|c} 1 & -1/3 & 0 & 0 \\ -2 & -4 & 2 & 0 \\ -4 & -8 & 4 & 0 \end{array}\right]$$

$R_2 - (-2) \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & -1/3 & 0 & 0 \\ 0 & -14/3 & 2 & 0 \\ -4 & -8 & 4 & 0 \end{array}\right]$$

$R_3 - (-4) \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & -1/3 & 0 & 0 \\ 0 & -14/3 & 2 & 0 \\ 0 & -28/3 & 4 & 0 \end{array}\right]$$

$R_2 \Big/ \left(\frac{-14}{3}\right)$

$$\sim \left[\begin{array}{ccc|c} 1 & -1/3 & 0 & 0 \\ 0 & 1 & -3/7 & 0 \\ 0 & -28/3 & 4 & 0 \end{array}\right]$$

$R_3 - \left(\frac{-28}{3}\right) \cdot R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & -1/3 & 0 & 0 \\ 0 & 1 & -3/7 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$R_1 - \left(\frac{-1}{3}\right) \cdot R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -1/7 & 0 \\ 0 & 1 & -3/7 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{aligned} x_1 - \frac{1}{7} \cdot x_3 &= 0 \\ x_2 - \frac{3}{7} \cdot x_3 &= 0 \end{aligned}\right\} \quad \text{... System (1)}$$

Find the variable x_2 from the equation 2 of the system (1) : $x_2 = \frac{1}{7} \cdot x_3$.

Find the variable x_1 from the equation 1 of the system (1) : $x_1 = \frac{1}{7} \cdot x_3$.

$$x_1 = \frac{1}{7} \cdot x_3$$

$$x_2 = \frac{3}{7} \cdot x_3$$

$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} \frac{1}{7} \cdot x_3 \\ \frac{3}{7} \cdot x_3 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_3 \cdot \begin{bmatrix} \frac{1}{7} \\ \frac{3}{7} \\ 1 \end{bmatrix} \right\}$

$$v_3 = \begin{bmatrix} \frac{1}{7} \\ \frac{3}{7} \\ 1 \end{bmatrix}$$

- The diagonal matrix (The diagonal entries are the eigenvalues - λ_1, λ_2 and λ_3 :

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

The matrix with the eigenvalues (v_1, v_2 and v_3) as its columns :

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{7} \\ \frac{1}{2} & \frac{1}{2} & \frac{3}{7} \\ 1 & 1 & 1 \end{bmatrix}$$

Ex. 1.7.3 : *Prove that* $A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$ *is diagonalizable and use it to find* A^{13}. **GTU : Summer-19, Marks 7**

Sol. : Then : $Av - \lambda v = (A - \lambda I) \cdot v = 0$

Equation has a non-zero solution if and only if

$$\det(A - \lambda I) = 0$$

$$\det(A - \lambda I) = \begin{vmatrix} 0-\lambda & 0 & -2 \\ 1 & 2-\lambda & 1 \\ 1 & 0 & 3-\lambda \end{vmatrix}$$

$$= -\lambda^3 + 5\lambda^2 - 8\lambda + 4$$

$$= -(\lambda - 1) \cdot (\lambda^2 - 4\lambda + 4)$$

$$= -(\lambda - 1) \cdot (\lambda - 2)^2 = 0$$

1. $\lambda_1 = 1$
2. $\lambda_2 = 2$

For every λ we find its own vectors :

1. $\lambda_1 = 1$

$$A - \lambda_1 I = \begin{bmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A - \lambda I) v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} -1 & 0 & -2 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right]$$

$R_1/(-1)$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right]$$

$R_2 - 1 \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right]$$

$R_3 - 1 \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{array}{r} x_1 + 2 \cdot x_3 = 0 \\ x_2 - x_3 = 0 \end{array}\right\} \quad \text{... System (1)}$$

Find the variable x_2 from the equation 2 of the system (1) : $x_2 = x_3$

Find the variable x_1 from the equation 1 of the system (1) : $x_1 = -2x_3$

$$x_1 = -2x_3$$
$$x_2 = x_3$$
$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} -2x_3 \\ x_3 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_3 \cdot \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$v_1 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 2$$

$$A - \lambda_2 I = \begin{bmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$Av = \lambda v^*$$

$$(A - \lambda I) \cdot v = 0$$

So we have a **homogeneous system** of linear equations, we solve it by Gaussian elimination :

$$\left[\begin{array}{ccc|c} -2 & 0 & -2 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{array}\right]$$

$R_1/(-2)$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{array}\right]$$

$R_2 - 1 \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{array}\right]$$

$R_3 - 1 \cdot R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$x_1 + x_3 = 0 \quad \text{... System (1)}$$

Find the variable x_1 from the equation 1 of the system (1) : $x_1 = -x_3$

$$x_1 = -x_3$$
$$x_2 = x_2$$
$$x_3 = x_3$$

General solution : $X = \begin{bmatrix} -x_3 \\ x_2 \\ x_3 \end{bmatrix}$

The solution set : $\left\{ x_2 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \cdot \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

The diagonal matrix (The diagonal entries are the eigenvalues - λ_1, λ_2 and λ_3) :

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

The **matrix with the eigenvectors** (v_1, v_2 and v_3) as its columns :

$$P = \begin{bmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$P^{(-1)} = \begin{bmatrix} -1 & 0 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$D = PAP^{-1}$$

$$\Rightarrow \quad A = PDP^{-1}$$

$$\Rightarrow \quad A^{13} = PD^{13}P^{-1}$$

$$= \begin{bmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}^{13} \begin{bmatrix} -1 & 0 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1^{13} & 0 & 0 \\ 0 & 2^{13} & 0 \\ 0 & 0 & 2^{13} \end{bmatrix} \begin{bmatrix} -1 & 0 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -8190 & 0 & -16382 \\ 8191 & 8192 & 8192 \\ 8191 & 0 & 16383 \end{bmatrix}$$

1.8 Cayley - Hamilton Theorem

Statement : Every square matrix satisfies its own characteristic equation.

Uses of Cayley-Hamilton theorem :

1) To calculate the positive integral powers of A.
2) To calculate the inverse of a square matrix A.

Ex. 1.8.1 : *Show that the matrix* $\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ *satisfies its own characteristic equation.*

Sol. : Let $A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$. The characteristic equation of A is $\lambda^2 - S_1\lambda + S_2 = 0$ where

S_1 = Sum of the main diagonal elements

$= 1 + 1 = 2$

$S_2 = |A| = 1 - (-4) = 5$

The characteristic equation is $\lambda^2 - 2\lambda + 5 = 0$

To prove $A^2 - 2A + 5I = 0$

$$A^2 = A(A) = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -4 \\ 4 & -3 \end{bmatrix}$$

$$A^2 - 2A + 5I = \begin{bmatrix} -3 & -4 \\ 4 & -3 \end{bmatrix} - \begin{bmatrix} 2 & -4 \\ 4 & 2 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

Therefore, the given matrix satisfies its own characteristic equation.

Ex. 1.8.2 : *If* $A = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$ *write* A^2 *in terms of A and I, using Cayley - Hamilton theorem.*

Sol. : Cayley - Hamilton theorem states that every square matrix satisfies its own characteristic equation.

The characteristic equation of A is $\lambda^2 - S_1\lambda + S_2 = 0$ where

S_1 = Sum of the main diagonal elements = 6

$S_2 = |A| = 5$

Therefore, the characteristic equation is $\lambda^2 - 6\lambda + 5 = 0$

By Cayley-Hamiltion theorem, $A^2 - 6A + 5I = 0$

i.e., $\quad A^2 = 6A - 5I$

Ex. 1.8.3 *Verify Cayley-Hamilton theorem, find* A^4 *and* A^{-1} *when* $A = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$.

Sol. : The characteristic equation of A is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$ where

$$S_1 = \text{Sum of the main diagonal elements} = 2+2+2=6$$

$$S_2 = \text{Sum of the minors of the main diagonal elements} = 3+2+3=8$$

$$S_3 = |A| = 2(4-1)+1(-2+1)+2(1-2) = 2(3)-1-2=3$$

Therefore, the characteristic equation is,

$$\lambda^3-6\lambda^2+8\lambda-3 = 0$$

To prove that : $A^3-6A^2+8A-3I=0$ ...(1)

$$A^2 = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}\begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix}$$

$$A^3 = A^2(A) = \begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix}\begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 29 & -28 & 38 \\ -22 & 23 & -28 \\ 22 & -22 & 29 \end{bmatrix}$$

$$A^3-6A^2+8A-3I = \begin{bmatrix} 29 & -28 & 38 \\ -22 & 23 & -28 \\ 22 & -22 & 29 \end{bmatrix} - \begin{bmatrix} 42 & -36 & 54 \\ -30 & 36 & -36 \\ 30 & -30 & 42 \end{bmatrix} + \begin{bmatrix} 16 & -8 & 16 \\ -8 & 16 & -8 \\ 8 & -8 & 16 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

To find A^4 :

$(1) \Rightarrow A^3-6A^2+8A-3I=0 \Rightarrow A^3 = 6A^2-8A+3I$ ...(2)

Multiply by A on both sides, $A^4 = 6A^3-8A^2+3A = 6(6A^2-8A+3I)-8A^2+3A$

$$\Rightarrow \quad A^4 = 36A^2-48A+18I-8A^2+3A = 28A^2-45A+18I$$

$$\Rightarrow \quad A^4 = 28\begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix} - 45\begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 18\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 196 & -168 & 252 \\ -140 & 168 & -168 \\ 140 & -140 & 196 \end{bmatrix} - \begin{bmatrix} 90 & -45 & 90 \\ -45 & 90 & -45 \\ 45 & -45 & 90 \end{bmatrix} + \begin{bmatrix} 18 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 18 \end{bmatrix} = \begin{bmatrix} 124 & -123 & 162 \\ -95 & 96 & -123 \\ 95 & -95 & 124 \end{bmatrix}$$

To find A^{-1} :

Multiplying (1) by A^{-1}, $A^2-6A+8I-3A^{-1}=0$

$$\Rightarrow \quad 3A^{-1} = A^2-6A+8I$$

$$\Rightarrow \quad 3A^{-1} = \begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix} - 6\begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 8\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix} - \begin{bmatrix} -12 & 6 & -12 \\ 6 & -12 & 6 \\ -6 & 6 & -12 \end{bmatrix} + \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix} = \begin{bmatrix} 3 & 0 & -3 \\ 1 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix}$$

$$\Rightarrow \quad A^{-1} = \frac{1}{3}\begin{bmatrix} 3 & 0 & -3 \\ 1 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix}$$

Ex. 1.8.4 : *Use Cayley-Hamilton theorem for the matrix* $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ *to find* $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$.

Sol. : Given $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. The characteristic equation of A is $\lambda^2 - S_1\lambda + S_2 = 0$ where

$$S_1 = \text{Sum of the main diagonal elements} = 1 + 3 = 4.$$

$$S_2 = |A| = 3 - 8 = -5$$

The characteristic equation is $\lambda^2 - 4\lambda^2 - 5 = 0$

By Cayley-Hamilton theorem, we get,

$$A^2 - 4A - 5I = 0 \qquad \ldots(1)$$

$$\begin{array}{r|l}
 & A^3 - 2A - 3I \\
\hline
A^2 - 4A - 5I & A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I \\
 & -A^5 - 4A^4 - 5A^3 \\
\hline
 & \quad -2A^3 + 11A^2 - A - 10I \\
 & \quad -2A^3 + 8A^2 + 10A \\
\hline
 & \qquad 3A^2 - 11A - 10I \\
 & \qquad 3A^2 - 12A - 15I \\
\hline
 & \qquad 0 + A + 5I
\end{array}$$

$$A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I = (A^2 - 4A - 5I)(A^3 - 2A + 3I) + A + 5I = 0 + A + 5I$$

$$= A + 5I = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} + 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 4 \\ 2 & 8 \end{bmatrix}$$

□□□